Factors Affecting College Attendance

Andrew Lord, Adamina Bilbrey, Nathan Gebremichael

Data 4030: Programming in R

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Research Questions

Our main research question for the project was: What variables related to socioeconomic status/privilege are predictors of a high school student's decision to attend college? Furthermore, can a student's average grades correspond to socioeconomic status? And if socioeconomic status is a predictor of college attendance, can we make any recommendations based on our model to address these predictors' influence?

Data Sources

The dataset we will be using is the college motivation dataset on D2L. This is a public dataset with information on academic performance, family background, and demographics with a measure of how likely it is for the student to attend college. It was made by the National Institute of Statistics and Geography (INEGI). It consists of data from over twenty-seven thousand high schools in Indonesia who graduated in 2014. This dataset was created to aid policymakers and educators in understanding what factors affect college attendance. There are eleven variables in the set: type of school (high school or vocational), school accreditation (A, B, or C where is the best and has higher accreditation), gender of student (male or female), interest in going to college, residence type (rural or urban), parents age, parents salary, house area (in m^2), average grades in high school for the student, if the parents went to college (yes or no), and the student's decision to attend college.

Key Variables

Out of these variables, we chose five to examine due to their relevance to our research questions.

- Parent salary- clearly tied to socioeconomic status. The most direct measure of a student's class of wealth we could find in the dataset.
- House area- A measure of how large the house a student lives in is and is very closely linked to socioeconomic status, as a bigger house typically indicates a wealthier family.
- Residence type- Urban homes are more closely linked to a higher income than rural, linking this variable to socioeconomic status and thus making it of interest to us.
- Gender- We chose gender as a focus as it often indicates a level of privilege in many countries worldwide.
- Average grades- Although not an overt sign of socioeconomic status like the other variables, we were interested in whether average grades could indicate socioeconomic status in this dataset.

Model

Creating our model was a very straightforward process. We first had to recode the college attendance variable as strictly binary. We transformed it from a variable with "True" or "False" as the levels to "1" or "0" respectively.

```
#Recode will_go_to_college as binary
college_motivation$will_go_to_college_binary <- ifelse(college_motivation$will_go_to_college == "True", 1, 0)
```

Then, we coded a proc glm statement for a logistic regression model with the binary college attendance variable as the dependent variable, and our previously discussed variables of interest (gender, residence type, parent salary, house area, and average grades) as the independent variables.

```
#Logistic regression model
college_motivation_model <- glm(will_go_to_college_binary ~ gender + residence + parent_salary + house_area + average_grades, data = college_motivation, family = "binomial")
summary(college_motivation_model)
```

This was both the first and last model we tried with the data, because it immediately turned out to be so strong and accurate.

Logistic Regression Results:

Variable	Estimate	Std. Error	z-value	p-value
Intercept	-44.249	3.298	-13.417	<2E-16
Gender	-0.254	0.194	-1.306	0.192
Residence	2.623	0.259	10.134	<2E-16
Parent Salary	1.22E-06	9.72E-08	12.54	<2E-16
House Area	0.103	0.009	11.176	<2E-16
Average Grades	0.333	0.034	9.665	<2E-16

Table 1:

The table above shows the factors from the dataset that we initially thought may be significant enough to include in our analysis. Looking at the gender variable, the p-value was significantly larger than our threshold of 0.05. This showed that gender did not hold many statistically significant implications, and therefore did not help us in predicting whether or not a student would attend college. The extremely low p-values of the residence, parent salary, house area, and average grades indicate that these variables are statistically significant in predicting the odds that a student will attend college. The quantitative variables area, parent salary, and average grades also all had a positive correlation, indicating that when these respective values increase, the likelihood of a student attending college increases as well. These predictors helped us create a model to understand the statistical relevance of the mentioned factors in students enrolling in universities.

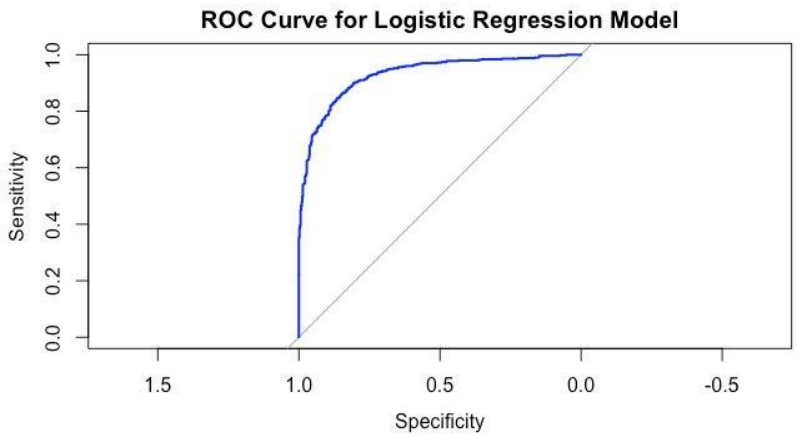


Figure 1: ROC Curve for Logistic Regression Model

The previous model showed how other factors affect college attendance, particularly the factors related to socioeconomic status. The main factor that appears to be affecting college attendance and does not look like a socioeconomic status factor is average grades. That is what we investigated next. A regression model was performed where the college attendance variable was dependent and average grades was independent. This gave results that showed that average grades have a statistically significant effect on the likelihood of going to college. The average grades estimate from the model was 0.42, which means that as average grade increases, the likelihood of going to college increases slightly. The p-value was found to be significantly lower than the threshold of significance of 0.5 and indicates that average grades influence the likelihood of college. Next, a prediction model was performed and visualized as figure 2 below. This shows how the variables are interacting and how clear it is from the models that average grades affect college attendance.

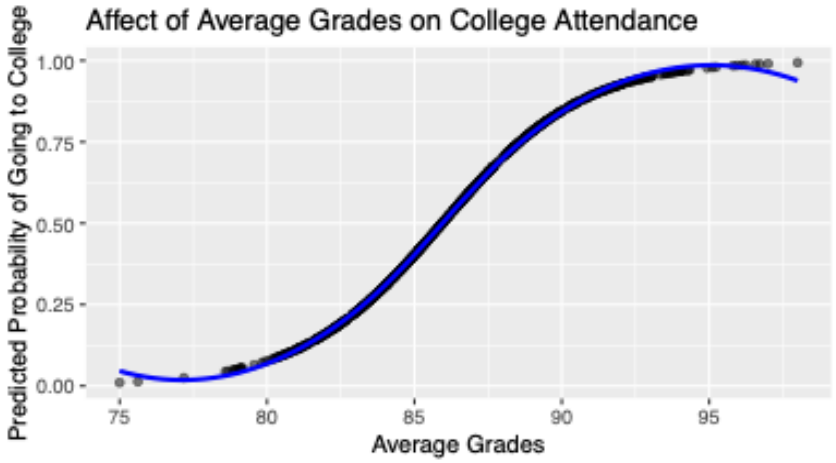


Figure 2: The Effect of Average Grades on College Attendance

Here we have the prediction model of how average grades affect colleges attendance for students. Once we determined that average grades do affect the likelihood of a student attending college, we investigate what other factors in the dataset may be affecting average grades of students. A regression model was performed that showed that parent salary and house area have the most highly significant p-values and thus affect average grades the most. Figures 3 and 4 visualize the effect of these factors on grades. This leads us to think that average grades, while potentially being affected by factors outside of the dataset, could be a socioeconomic factor as well.

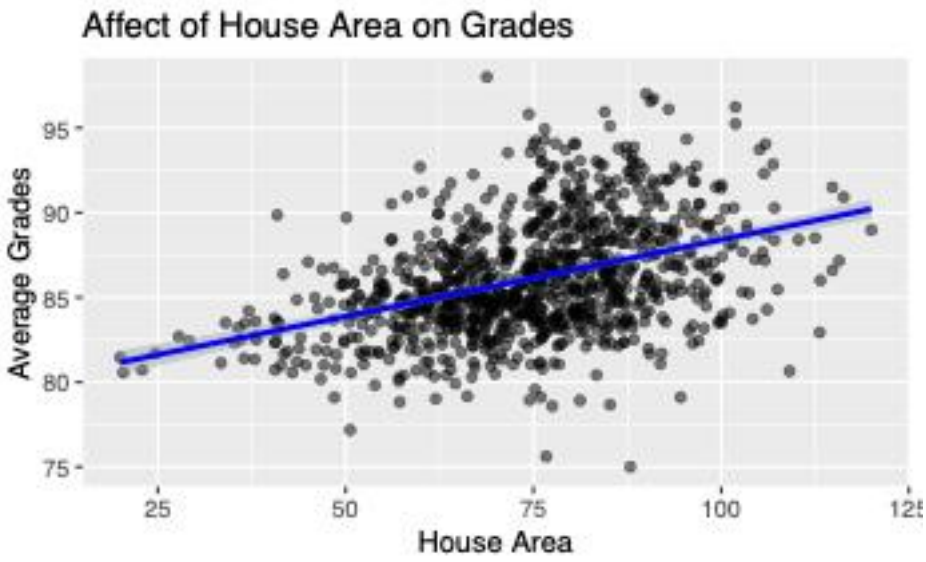


Figure 3: Scatterplot of the Effect of House Area on Grades

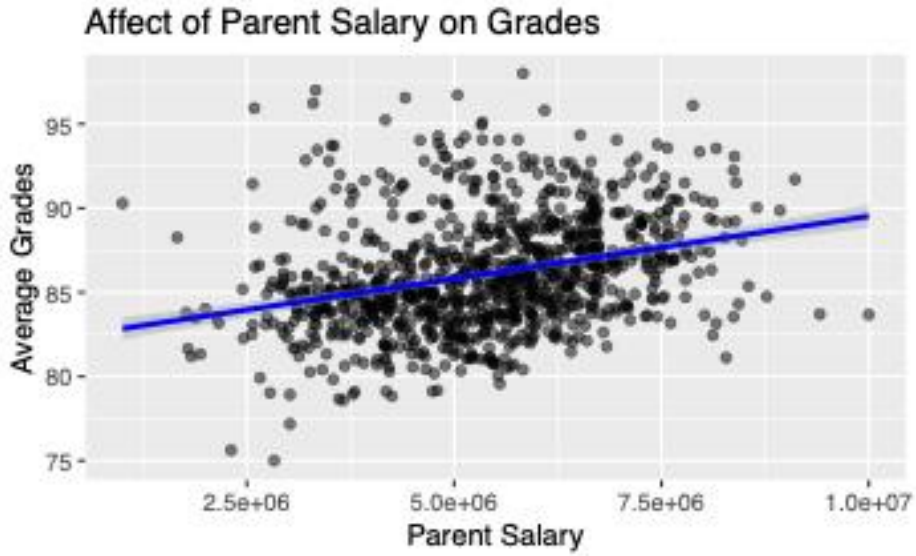


Figure 4. Scatterplot of the Effect of Parent Salary on Grades

USA Comparison

Since the data we used in this project is about students in Indonesia, we wanted to see how our results correlate to high school students in the USA, if at all. First, we found a report that found that in 2022, 18–24-year-olds in different family income quartiles, would have different percentages who went to college. The lowest quartile was at forty-four percent, while the highest quartile was seventy-nine percent. Overall, they found that "historical data suggests that higher levels of educational attainment are associated with higher incomes and greater job opportunities" (Hamm, 2024). Additionally, another report had similar findings, which are shown in figure 5 (Reber and Smith, 2023). This bar chart shows that has a socioeconomic status quartile level increase, the percentage of students who would go to college also increased. These two reports show that there is a significant wealth and college connection and disparity happening in the USA. Therefore, there is some correlation between the data in Indonesia and what is being seen in the USA. This is an area for more research and an area we did not find good raw data for the sake of this project to further inspect.

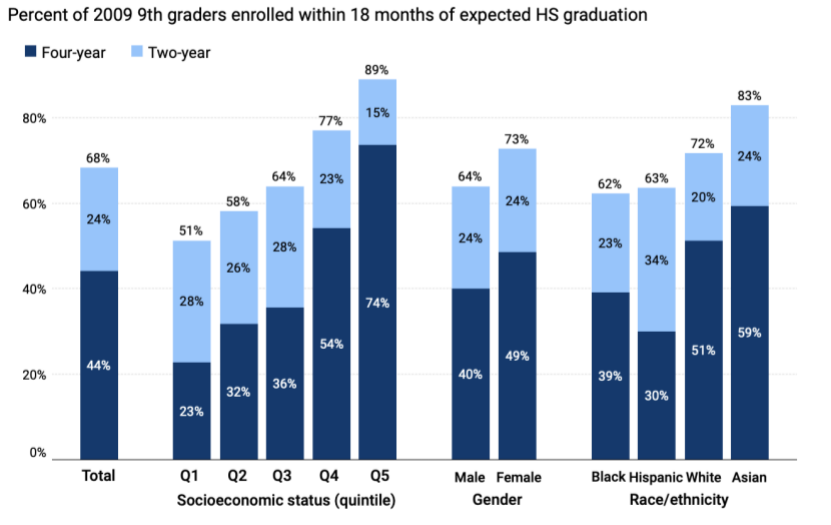


Figure 5: Post-secondary Enrollment Rates (Reber and Smith)

Discussion

Our overall model's results clearly show that socioeconomic status and privilege are influential on Indonesian high school students' decision to attend college. This reflects general global trends as well as trends here in the United States.

More specifically, the results show that a parent having a greater salary means that a student is significantly likelier to attend college. Thus, it stands to reason that this is because the student has significantly more financial support from their parents, making them more equipped to deal with college's tuition/fees and to pay off student debt. Based on this finding, we recommend that Indonesian colleges should reduce tuition/fees, make financial aid more widely available for students, and introduce widespread student debt forgiveness. The high financial costs of college make the disparity between students with higher income parents and students with lower income parents inevitable and should be addressed.

Additionally, we found that average grades are a potential identifier of socioeconomic status or privilege. In other words, students that belong to families with greater wealth have

access to more resources, such as better tutoring and quality private schools, that increase their grades. On the other hand, students with access to less wealth may have more responsibilities that keep them from devoting as much time to school, thus leading to lower grades. Based on these results, we recommend that Indonesian high schools implement some way to identify students whose financial struggles may be leaking into their grades, and offer them free, effective, and extra help tailored to their needs.

There were multiple limitations to our analysis. Firstly, our model, despite appearing quite accurate, may be compromised due to our failure to check it for multicollinearity. We also wanted to examine race/ethnicity as a factor in relation to socioeconomic status/privilege, but our dataset did not measure it. Moreover, with more time, we would have tried other models, specifically using test and train subsets of the data. Finally, we would have incorporated an additional raw data set from the USA to run our own analysis on and compare to the Indonesian data, but due to time constraints, we had to settle for only looking at existing analysis/trends from the USA.

References

Hexter, Holly, et al. “New Pell Institute Report Shows Decline in the Global Position of the United States in Bachelor’s Attainment, Increasing Inequality of College Opportunity at Home.” Council for Opportunity in Education, 10 May 2024, coenet.org/news-impact/press-releases/new-pell-institute-report-shows-decline-in-the-global-position-of-the-united-states-in-bachelors-attainment-increasing-inequality-of-college-opportunity-at-home/#:~:text=The%20report%20indicates%20that%20in,from%20the%20lowest%20income%20quartile.

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